

Feb 25: polynomials.

1. Find the quotient and remainder of $\frac{f(x)}{g(x)}$

$$f(x) = x^4 - 3x^2 - 2x - 1, \quad g(x) = x^2 - 2x - 5$$

2. (Chinese remainder theorem) Let $g_1, \dots, g_n(x)$ be coprime polys.

$r_1(x), \dots, r_n(x)$ be arbitrary n polys. prove that there exist a poly $f(x)$ s.t. $f(x) = g_i(x) p_i(x) + r_i(x) \quad (1 \leq i \leq n)$

3. Let $f(x)$ be a deg n poly in $\mathbb{Z}[x]$, and $\exists$ distinct integers $a_1, \dots, a_{n+1}$ s.t. $|f(a_i)| = 1 \quad \forall i$, prove that $f(x)$ is irreducible in $\mathbb{Q}[x]$.

4. Let p be a prime number. prove that the polynomial

$$f(x) = (p-1)x^{p-2} + (p-2)x^{p-3} + \dots + 2x + 1 \text{ is irreducible in } \mathbb{Q}[x].$$

5. V vector space / K . $g: V \rightarrow V$. Let $f(x), g(x) \in K[x]$ polys

s.t. $d(x) = (f(x), g(x))$. $m(x) = [f(x), g(x)]$, are Lcd and Lcm

please prove that $\ker d(\varphi) = \ker f(\varphi) \cap \ker g(\varphi)$. $\ker m(\varphi) = \ker f(\varphi) + \ker g(\varphi)$

6. Let $T: U \rightarrow V$ be a linear map. and suppose that

$$\dim(U) = \dim(V) = n. \quad \text{The TFAE}$$

i) T is surjective

ii) $\text{rank}(T) = n$

iii) $\text{nullity}(T) = 0$

iv) T is injective.

7. Let $a_1, \dots, a_n \in \mathbb{F}$ be n distinct. Show that $f_j(x) = \prod_{i \neq j} (x - a_i)$ form a basis of $\mathbb{F}[x]_{n-1} := \{f(x) \in \mathbb{F}[x], \deg f(x) \leq n-1\}$

8. Assume real number series set $S := \{(x_1, x_2, \dots, x_n, \dots) \mid x_n = x_{n-2} + x_{n-1} \text{ for } n \geq 3\}$ and S form a linear space / $\mathbb{R}$ by

$$+'' \quad (x_1, \dots, x_n, \dots) + (y_1, \dots, y_n, \dots) = (x_1 + y_1, \dots, x_n + y_n, \dots)$$

$$\text{Scalar multiplication.} \quad k(x_1, \dots, x_n) = (kx_1, \dots, kx_n)$$

(1) Find a basis of S and find $\dim(S)$

(2) Find a basis of S consisting some geometric series

(3). Find the general term formula F_n of the Fibonacci series

$$(0, 1, 1, 2, 3, 5, \dots)$$

Soln 2 We first find polys $f_i(x)$ s.t

$$f_i(x) = g_i(x) p_i(x) + 1 \quad \forall i \text{ and } g_j(x) \nmid f_i(x) \quad \forall j \neq i$$

Once we have those $f_i(x)$. Let

$$f(x) = r_1(x) f_1(x) + \dots + r_n(x) f_n(x). \quad \text{then we are done.}$$

To construct $f_i(x)$.

$$(i=1) \quad \text{since } |g_1(x), g_j(x)| = 1 \text{ for } j \neq 1. \Rightarrow \exists u_j(x), v_j(x) \text{ s.t.}$$

$$g_1(x) u_j(x) + g_j(x) v_j(x) = 1$$

$$\text{Let } f_1(x) = g_2(x) v_2(x) g_3(x) v_3(x) \dots g_n(x) v_n(x)$$

$$= (1 - g_1(x) u_2(x)) (1 - g_1(x) u_3(x)) \dots (1 - g_1(x) u_n(x)) \quad \square$$

Similarly for other $f_i(x)$

3. By contradiction. Assume $f(x) = g(x) h(x)$ in $\mathbb{Q}[x]$. then we must have $g(x), h(x) \in \mathbb{Z}[x]$. (By Gauss-Lemma)

W.L.O.G. we assume $\deg g(x) \leq \lfloor \frac{n}{2} \rfloor$ when $n = \deg f(x)$

$$\text{Since } f(a_i) = \pm 1 \Rightarrow g(a_i) = \pm 1$$

$$\text{i.e. } g^2(a_i) - 1 = 0 \quad \forall i.$$

$$\Rightarrow \deg 2 \lfloor \frac{n}{2} \rfloor \leq n \quad \text{poly } g^2(x) - 1 \text{ has distinct } n+1 \text{ roots } a_1, \dots, a_{n+1}.$$

Contradiction.

$$4. \text{ Notice that } f(x) = (x^{p-1} + x^{p-2} + \dots + x^1 + x^0)' = \left(\frac{x^p - 1}{x - 1} \right)'$$

$$\begin{aligned} \cdot \text{ Let } x = y+1, \text{ then } f(x) &= g(y) = \frac{d}{dy} \left(\frac{(y+1)^p - 1}{y} \right) \\ &= (p-1)y^{p-2} + (p-2)C_p^1 y^{p-3} + \dots + 2C_p^{p-3} + C_p^{p-2} \end{aligned}$$

By Eisenstein criterion $\Rightarrow g(y)$ is irreducible in $\mathbb{Q}[y]$.

5. Since $(f(x), g(x)) = d(x)$. $[f(x), g(x)] = m(x)$. $\Rightarrow$

$$\textcircled{1} f(x) = f_1(x) d(x), \quad g(x) = g_1(x) d(x) \quad \text{and} \quad \textcircled{2} m(x) = f_1(x) g_1(x) d(x).$$

$$\textcircled{3} \exists u(x), v(x) \text{ s.t. } f_1(x) u(x) + g_1(x) v(x) = 1$$

$$(1) \text{ By } \textcircled{1} \quad \ker d(\varphi) \subseteq \ker f(\varphi) \quad \ker d(\varphi) \subseteq \ker g(\varphi)$$

$$\Rightarrow \ker d(\varphi) \subseteq \ker f(\varphi) \cap \ker g(\varphi)$$

$$\text{on the other hand. since } d(x) = f_1(x) u(x) + g_1(x) v(x)$$

$$\Rightarrow \forall v \in \ker(f(\varphi) \cap \ker g(\varphi)$$

$$\text{we have } d(\varphi)(v) = u(\varphi) f(\varphi)(v) + v(\varphi) g(\varphi)(v)$$

$$\Rightarrow d(\varphi)(v) = 0 \Rightarrow v \in \ker d(\varphi)$$

$$(2) \text{ By } \textcircled{2} \quad \ker f(\varphi) \subseteq \ker m(\varphi), \quad \ker g(\varphi) \subseteq \ker m(\varphi)$$

$$\Rightarrow \ker f(\varphi) + \ker g(\varphi) \subseteq \ker m(\varphi)$$

$$\text{on the other hand } \forall w \in \ker m(\varphi)$$

$$w = \underbrace{f_1(\varphi) u(\varphi)(w)}_{\in \ker g(\varphi)} + \underbrace{g_1(\varphi) v(\varphi)(w)}_{\in \ker f(\varphi)}$$

□

6. By $\text{rank}(T) + \text{nullity}(T) = \dim V$.

7. Only need to prove that $f_i(x)$ are linearly indep.

$$\text{Suppose } c_1 f_1(x) + \dots + c_n f_n(x) = 0$$

$$\text{substitute } a_i \Rightarrow c_1 f(a_i) = 0 \Rightarrow c_i = 0$$

$$\begin{aligned}
 8.1) \forall \alpha &= (\alpha_1, \alpha_2, \alpha_1 + \alpha_2, \alpha_1 + 2\alpha_2, 2\alpha_1 + 3\alpha_2, \dots) \\
 &= \alpha_1 (1, 0, 1, 1, 2, 3, 5, \dots) \\
 &\quad + \alpha_2 (0, 1, 1, 2, 3, 5, \dots) = \alpha_1 \xi_1 + \alpha_2 \xi_2.
 \end{aligned}$$

clearly ξ_1 and ξ_2 are linearly indep $\Rightarrow \dim S = 2$

2) Assume a geometric series

$$(0, aq, aq^2, aq^3, \dots) \in S$$

$$\text{then } aq^n = aq^{n-2} + aq^{n-1} \Rightarrow q^2 = 1 + q \Rightarrow q_1 = \frac{1+\sqrt{5}}{2}, q_2 = \frac{1-\sqrt{5}}{2}.$$

$$\Rightarrow \eta_1 = (1, q_1, q_1^2, \dots)$$

are linearly indep and hence form a basis

$$\eta_2 = (1, q_2, q_2^2, \dots)$$

3) Use the basis $\{\eta_1, \eta_2\}$

$$(0, 1, 1, 2, 3, \dots) = c_1 \eta_1 + c_2 \eta_2$$

$$\Rightarrow c_1 + c_2 = 0 \quad c_1 q_1 + c_2 q_2 = 1$$

$$\Rightarrow c_1 = \frac{\sqrt{5}}{5} \quad c_2 = -\frac{\sqrt{5}}{5}$$